

Taslak Makale

Halil İbrahim Kaplan¹[0009–0006–0120–6512]

Cryptanalysis Laboratory, TÜBİTAK BİLGEM, Kocaeli, Turkey
halil.kaplan@tubitak.gov.tr

Abstract. Abstact buraya gelecek

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1 Introduction

Cryptanalysis of block ciphers has long been a cornerstone of cryptographic research, driven by the need to ensure

Organization of the paper. The rest of the paper is organized as follows. Section 2 introduces Linear and mixed-integer linear programming with example. Section 3 defines ITUbee algorithm. Section 4 studies how to model block cipher components and. Section 6 introduces our MILP models for differential, linear, relatekey differential cryptanalysis

2 LP and MILP

2.1 Linear Programming

Linear Programming (LP) is a fundamental mathematical optimization technique that was first introduced in the 1940s by George Dantzig [1], although its development continued into the 1950s. LP has become an

Modelling and Result We will use Sagemath and GLPK solver for modeling Toy example. After running code in Listing 1.1 . We found that Maximum profit 2160 can be achieved by producing 42 A product and 16 B product.

Listing 1.1. Sagemath MILP modelling of toy example

```
1 p.<x,y> = MixedIntegerLinearProgram
2 (maximization=True, solver = "GLPK")
3
4
5 p.set_objective(40*x[0]+ 30*y[0])
6
7 p.add_constraint(2*x[0] + y[0] == 100)
8 p.add_constraint(x[0] + 3*y[0] == 90)
```

```
9 p.add_constraint(x[0] >= 20)
10 p.add_constraint(y[0] >= 0)
11
12 print(p.solve())
13 print(p.get_values(x,y))
```

3 Conclusion

As a conclusion, our results demonstrate that ITUbee exhibits strong resistance to classical differential and linear cryptanalysis for more than three rounds, in alignment with the theoretical claims of the original authors. No effective differential or linear trails with sufficiently high probability or correlation could be found beyond this threshold, affirming the cipher's robustness in this context.

Overall, this study reinforces the value of MILP as a precise, automatable, and interpretable method for evaluating security of cryptographic algorithms. All MILP models we developed can be found at <https://github.com/KaplanHalil/Milp/tree/main/ITUBEE> and discovered characteristics are included within the thesis to support further research and reproducibility.

References

1. Dantzig, G.B.: Linear programming. Operations research **50**(1), 42–47 (2002)